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CHAIR FOR THE SITTING OF A USER AND METHOD OF MOUNTING SAID CHAIR

Abstract

Chair comprising a support structure provided with a first housing seat having a first non-circular opening, and with a support element, which is mechanically connected to the support structure and is provided with a second housing seat, which is aligned and communicating with the first housing seat. The chair also comprises a removable connection system having a locking pin provided with a non-circular head, which engages the first housing seat and the second housing seat and is rotatable between a release position, in which it releases the support element from the support structure, and a constraint position, in which the support element is constrained to the support structure. The locking pin comprises an elastic retention element which is elastically engaged in a first retention recess of the first housing seat when the locking pin is in the constraint position, in order to retain the locking pin therein.

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Background/Summary

FIELD OF APPLICATION

[0001] The present invention regards a chair for the sitting of a user and a method of mounting said chair.

[0002] The present chair is intended to be employed for allowing the sitting of a user, both for example in offices, studies and the like (in particular during working hours) and in private homes, for example in the respective studies, kitchens or living rooms.

[0003] The present chair is advantageously of reclinable type, so as to allow a user to vary his/her own sitting position between a first position in which the back is placed substantially at a right angle with respect to the legs, and multiple reclined positions.

[0004] The chair, object of the present invention, and the relative mounting method, are therefore inserted in the field of furnishing settings, both work and home and more generally in the field of furniture production, in particular in the production of chairs and reclinable chairs.

STATE OF THE ART

[0005] The chairs of known type comprise a support base, which is intended to be rested on the floor, a seat, which is mounted on the support base and is intended to support a user in seated position, a back and possible arms.

[0006] The chairs of known type can be made in a single body, e.g. of plastic or metal, or alternatively they can be formed by multiple separate elements which are connected to each other in the chair mounting step.

[0007] A chair in a single body, also termed unibody, is usually a product that is less expensive and with lower added value, and is often intended to be employed for a temporary, not extended use of the chair, since it has limited ergonomics and comfort, unsuitable for a use of the chair for extended time periods and in more professional settings, for example as an office chair.

[0008] Otherwise, the chairs suitable for more professional uses (also termed ergonomic), and which have greater comfort and ergonomics, are formed by multiple separate elements that must be subsequently assembled together, before proceeding with the shipping and with the sale at the retailer or at the final client.

[0009] Usually, such ergonomic chairs comprise a support base, which can be made of metal or plastic, on which a seat is mounted that is usually formed by a rigid support covered with a padding layer or, alternatively, by a mesh support that likewise allows good user comfort.

[0010] In addition, such ergonomic chairs have a back, in order to support the user's back in seated position, and a pair of arms adapted to allow the user to rest his/her arms during the use of the chair.

[0011] The back is connected to the support base or alternatively to the seat, and is in particular hinged to the base or to the seat and reclinable between a raised position and a reclined position, in order to increase the ergonomics of the chair during use depending on the client needs.

[0012] The arms are instead fixed to the support base or to the seat, and can have a telescopic portion that allows adjusting the height of the arms in order to increase the ergonomics of the chair depending on the needs of the client.

[0013] In the ergonomic chairs of known type, the elements that make up the chair are fixed together by means of screws and are normally preassembled before proceeding with the shipping of the chair to the supplier or to the final client.

[0014] The above-described chairs of known type, both in the unibody version and in the

ergonomic version, have proven in practice that they do not lack drawbacks.
[0015] The main drawback of the aforesaid chairs of known type is given by the fact that they have proven bulky and uncomfortable, as well as costly, to transport.
[0016] Indeed, for various reasons, both of the aforesaid types of chairs of known type require that the chair be sent to the retailer or to the final consumer already assembled, and hence with the back and the arms fixed to the seat or to the support base, determining higher transport costs for the chairs by the producer.
[0017] In addition, a further drawback lies in the fact that such assembly step determines an increase of the costs and times for producing the chairs, since it requires a suitable step in which the back and the arms are fixed to the seat or to the support base, by means of screws, bolts or alternatively by means of welding of metal elements.
[0018] A further drawback lies in the fact that such chairs of known type have proven incompatible for allowing a readaptation thereof, both of the number of elements and of their configuration. Indeed, the assembly of the chairs of known type is usually irreversible, in particular with regard to possible welds, and it is in any case complicated to disassemble and remount the chair in the event in which the user wishes to modify the structure of the chair, or even only to facilitate its transport (e.g. in moves/relocations).
PRESENTATION OF THE INVENTION
[0019] In this situation, the problem underlying the present invention is therefore that of overcoming the drawbacks manifested by the support elements for chairs of known type, by providing a chair for the sitting of a user which is simple and inexpensive to transport.
[0020] A further object of the present invention is to provide a chair which is simple and easy to mount. A further object of the present invention is to provide a chair which is simple and easy to disassemble, also by the final user and without particular operations.
[0021] A further object of the present invention is to provide a chair which can be readapted in different configurations depending on the user needs.
[0022] A further object of the present invention is to provide a method of mounting a chair which is quick and easy to execute, also by non-qualified personnel.
[0023] A further object of the present invention is to provide a chair which allows a reduction of the steps necessary for placing it in working condition.
[0024] A further object of the present invention is to provide a chair which is safe and stable during the use thereof.
[0025] A further object of the present invention is to provide a chair which is aesthetically appreciable.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0026] The technical characteristics of the invention, according to the aforesaid objects, and the advantages thereof, will be more evident in the following detailed description, made with reference to the enclosed drawings, which represent a merely exemplifying and non-limiting embodiment of the invention, in which:
[0027] FIG. **1** shows a perspective view of the chair for the sitting of a user, object of the present invention;
[0028] FIG. **2** shows a perspective view of a detail of the chair of FIG. **1**, relative to a locking pin in accordance with a first embodiment;
[0029] FIG. **3** shows a further perspective view of the locking pin of FIG. **2**;
[0030] FIG. **4** shows a perspective view of a detail of the chair of FIG. **1**, relative to a locking pin in accordance with a second embodiment;

[0031] FIG. **5** shows a further perspective view of the locking pin of FIG. **4**;
[0032] FIG. **6** shows a sectional view of a detail of the chair of FIG. **1**, with the locking pin of FIG. **2** in an insertion position;
[0033] FIG. **7** shows a sectional view of a detail of the chair of FIG. **1**, with the locking pin of FIG. **2** in a release position;
[0034] FIG. **8** shows a top view of the chair of FIG. **1**, with the locking pin in an insertion position, and with several parts removed in order to better illustrate other parts;
[0035] FIG. **9** shows a top view of the chair of FIG. **1**, with the locking pin in a release position, and with several parts removed in order to better illustrate other parts;
[0036] FIG. **10** shows a sectional view of the detail of FIG. **8**, with the locking pin in an insertion position, and with several parts removed in order to better illustrate other parts;
[0037] FIG. **11** shows a sectional view of the detail of FIG. **9**, with the locking pin in a release position, and with several parts removed in order to better illustrate other parts;
[0038] FIG. **12** shows a further view of the detail of FIG. **11**, made from a different section;
[0039] FIG. **13** shows a sectional view of a detail of the chair of FIG. **1**, with the locking pin in a constraint position;
[0040] FIG. **14** shows a sectional view of the detail of FIG. **13**, with the locking pin in a constraint position, and with several parts removed in order to better illustrate other parts;
[0041] FIG. **15** shows a further view of the detail of FIG. **14**, made from a different section;
[0042] FIG. **16** shows a further view of the detail of FIG. **13**, made from a different section;
[0043] FIG. **17** shows a rear view of the locking pin of FIG. **2**;
[0044] FIGS. **18**, **19** and **20** show a sectional view of the chair of FIG. **1**, with the locking pin respectively in a release position, insertion position and constraint position.

DETAILED DESCRIPTION OF A PREFERRED EMBODIMENT

[0045] With reference to the enclosed drawings, reference number **1** overall indicates chair for the sitting of a user, object of the present invention.
[0046] Advantageously, the present chair **1** is adapted to be placed as a furnishing element in both public and private settings, both work and home environments.
[0047] More in detail, the chair **1**, object of the present invention, is intended to be advantageously used for the sitting of a user, for example for the sitting of a worker in an office or for the sitting of a user in a private and/or home setting.
[0048] In particular, the chair **1**, object of the present invention, is advantageously of reclinable type, so as to allow a user to vary his/her own sitting position between a first position, e.g. an operative position in which the back is placed substantially at right angle with respect to the legs, and multiple reclined positions, substantially rest positions, in which the back is tilted with respect to the legs and the user takes on a more laid back position with respect to the aforesaid operative position.
[0049] The chair **1** for the sitting of a user according to the invention comprises a support structure **2**, provided with a support base **21**, which is intended to be placed in abutment against the ground (e.g. a floor).
[0050] Advantageously, the support base **21** of the chair **1** comprises multiple support legs which are preferably provided with wheels in order to allow the movement of the chair on the floor during its use.
[0051] Of course, it is also possible to provide for a support base **21** of base type, without support legs, or even provide for a support base **21** in which the legs lack wheels, so as to allow greater stability of the chair **1** in position during the use thereof.
[0052] Preferably, the support base **21** of the chair **1** is made of resistant material, in particular of metal material, such as for example steel or aluminum, or alternatively of a plastic material.
[0053] The chair **1** also comprises a seat **210**, mounted above the support base **21** and intended to support a user in seated position. Without departing from the protective scope of the present

invention, the seat **210** can be fixed with respect to the support base **21** or, alternatively, rotatable and preferably adjustable, e.g. height-wise, with respect to the same support base **21**.

[0054] In particular, the seat **210** defines a seat plane, substantially horizontal, and it can be formed by a rigid element (in a single body) or by a perimeter frame to which a support element (e.g. a mesh) is fixed in order to allow a certain flexibility (and ergonomics) of the seat. Alternatively, it is also possible that the seat **210** comprises a padding layer, placed on top of the rigid element, which is subsequently covered by a cover (made of leather, imitation leather or fabric) which encloses the rigid element and the padding layer.

[0055] In accordance with the invention, the support structure **2** is provided with at least one first housing seat **20**, which is provided with a first opening **200**, preferably non-circular.

[0056] More in detail, as is better explained hereinbelow, such first housing seat **20** can be made on the support base **21** or on the seat **210**, depending on the desired configuration and design of the chair **1**.

[0057] More in detail, the first opening **200** is accessible externally, and is preferably placed in a position in which it can simultaneously be hidden from view but easily accessible by the user, preferably at one face of the support structure **2** that is directed towards the floor during the normal use of the chair **1**.

[0058] Advantageously, the first housing seat **20** is a through seat, which is extended between the first opening **200** and an outlet opening **201**, which preferably also has a non-circular shape.

[0059] The chair **1** also comprises at least one support element **3**, which is mechanically connected to the support structure **2** and is selected from among a back **31** and a pair of arms **32**, in order to respectively support the back or the arm of the user, in which the support element **3** is provided with a second housing seat **30**, which is aligned and communicating with the first housing seat **20** of the support structure **2**, when the support element **3** is mounted on the support structure **2**.

[0060] The support element **3** can be connected, depending on the needs and on the configuration of the chair **1**, to the support base **21** or to the seat **210**, without departing from the protective scope of the present invention.

[0061] Advantageously, in a non-limiting manner, in the event in which the support element **3** is a back **31**, this is preferably connected to the support base **21**, while the pair of arms **32** is preferably connected to the seat **210**.

[0062] More in detail, the second housing seat **30** is provided with a second opening **300**, preferably also with non-circular shape, which is aligned (and preferably coinciding) with the outlet opening **201** of the first housing seat **20**.

[0063] Preferably, the second housing seat **30** is not through, and is only open at the second opening **300**, otherwise being blind in order to prevent the access also from an opposite side with respect to the second opening **300**.

[0064] In other words, when the support element **3** is mounted on the support structure **2**, the first housing seat **20** and the second housing seat **30** are aligned with each other and define a single housing seat which is accessible, preferably exclusively, from the first opening **200**.

[0065] The chair **1** also comprises a removable connection system **4**, which is placed as a mechanical connection between the support structure **2** and the support element **3**.

[0066] More in detail, with the expression removable connection system **4** a system is intended which allows a facilitated mounting and disassembly of the chair **1** and of the relative components, also by a user, even during the lifetime of the chair and not simply following the production process of the chair **1** itself. In other words, as is clearer below, it is not only intended a system that if removed allows the separation of the components of the chair **1**, but also means that it facilitates such separation.

[0067] The removable connection system **4** comprises a locking pin **41**, which is provided with a non-circular head **410** and which engages the first housing seat **20** and the second housing seat **30**. In particular, the locking pin **41** engages, at least with its own head **410**, the first housing seat **20** and the second housing seat **30**.

[0068] More in detail, the head **410** is provided with a form which has a radial asymmetry, and is preferably provided with an elongated form, substantially rectangular, such to determine a plurality of distinct positions during its rotation around the rotation axis R, as is better explained hereinbelow. In particular, the head **410** is extended along a direction substantially orthogonal to the rotation axis R.

[0069] Advantageously, the first opening **200** of the first engagement seat **20** is counter-shaped with respect to the head **410**, and still more preferably also the outlet opening **201** and the second opening **300** are counter-shaped with respect to the head **410**. In this manner, the locking pin **41** can be inserted with its own head **410** in an unambiguous manner, with the head **410** positioned in a specific insertion position A.

[0070] In accordance with a first embodiment of the locking pin **41**, illustrated in FIGS. **2** and **3**, the head **410** of the pin **41** is provided with a tooth **419**, which projects above the pin **41**, which is intended to abut against a shoulder made in the second housing seat **30**, in particular in the second housing seat **30** of the pair of arms **32**, in order to maintain the arm in position, preventing the removal thereof.

[0071] Otherwise, in accordance with a second embodiment of the locking pin **41**, illustrated in FIGS. **4** and **5**, the head **410** is flat, and therefore preferably has an upper surface that is substantially planar.

[0072] The locking pin **41** is rotatable in a constraint sense, around a rotation axis R, between a release position B, in which the locking pin **41** releases the support element **3** from the support structure **2**, and a constraint position C, in which the locking pin **41** constrains the support element **3** to the support structure **2**.

[0073] In accordance with the idea underlying the present invention, the locking pin **41** of the removable connection system **4** comprises an elastic retention element **411**, and the support structure **2** defines, in the first housing seat **20**, at least one first retention recess **21′**, in which the retention element **411** is elastically engaged with the locking pin **41** in release position B, in order to retain the locking pin **41** in position, by impeding the rotation thereof around the rotation axis R.

[0074] More in detail, the first retention recess **21′** is positioned such that the elastic retention element **411** is engaged in the latter when the locking pin **41** is situated in the release position B.

[0075] Advantageously, the elastic retention element **411** is configured in order to be elastically deformed along a direction substantially orthogonal to the rotation axis R during the rotation of the locking pin **41** around the rotation axis R, and return into a rest position when it reaches the first retention recess **21′**.

[0076] In this manner, it is possible to easily and quickly mount the support element **3** on the support structure **2**, by means of a simple rotation of the locking pin **41**, while simultaneously ensuring the safe and stable positioning of the locking pin **41** in the constraint position, reducing the risk of undesired movements of the latter which could determine the release of the support element **3** from the support structure **2**.

[0077] Advantageously, the support structure **2** defines, in the first housing seat **20**, also a second retention recess **21″**, which is engageable by the elastic retention element **411**, with the locking pin **41** in the constraint position C, in order to retain the locking pin **41** in the aforesaid constraint position C.

[0078] Advantageously, the first retention recess **21′** and the second retention recess **21″** are made on opposite sides of the first housing seat **20**, substantially spaced by an angle equal to 180°. In other words, the locking pin **41** is advantageously rotated 180° in order to pass from the release position B to the constraint position C, and vice versa, so as to prevent a small rotation of the locking pin **41** from determining the release of the support element **3** from the support structure **2**.

[0079] Advantageously, the locking pin **41** comprises a main body **413**, substantially cylindrical, which is provided with a lateral surface.

[0080] Preferably, the main body **413** is substantially housed entirely within the first housing seat **20** of the support structure **2**.
[0081] Advantageously, the locking pin **41** comprises a grip portion **415**, preferably placed outside the first housing seat **20** of the support structure **2**, and which is advantageously provided with an enlarged form.
[0082] Advantageously, the grip portion **415** is placed to close the first opening **200** of the first housing seat **20** and defines a surface that is substantially continuous with the support structure **2**, in order to reduce the aesthetic impact of the locking pin **41**.
[0083] Advantageously, the entire locking pin **41** is extended along a main extension direction X between the grip portion **415** and the head **410**, with the main body **413** that connects together the grip portion **415** and the head **410**, as is visible in FIGS. **2-5**.
[0084] Preferably, the main body **413** is extended along the aforesaid main extension direction X, substantially parallel to the rotation axis R, between a lower end, connected to the grip portion **415**, and an opposite upper end, connected to the head **410**.
[0085] More in detail, the head **410** is extended projecting from the main body **413**, along a direction substantially orthogonal to the main extension direction X.
[0086] In other words, the locking pin **41** has a substantially L shape.
[0087] Advantageously, the elastic retention element **411** is extended radially projecting from the lateral surface of the main body **413**. Preferably, the elastic retention element **411** is extended along a direction Y that is substantially orthogonal to the main extension direction X of the main body **413**.
[0088] More in detail, the elastic retention element **411** is susceptible of being elastically deformed along the direction Y, in particular being compressed during the rotation of the locking pin **41** around the rotation axis R, and on the contrary returning to a rest position, substantially not compressed, when the locking pin **41** reaches one of the retention recesses **21′**, **21″**.
[0089] Advantageously, the support structure **2** is provided with an internal surface **203**, which is directed towards the interior of the first retention seat **20** and laterally delimits the first retention seat **20**, and the retention recesses **21′**, **21″** are made in a depression from the internal surface **203** of the support structure **2**, along a radial extension direction. More in detail, the retention recesses **21′**, **21″** are extended in a depression along a direction parallel to the direction Y and substantially orthogonal to the rotation axis R. Preferably, the retention recesses **21′**, **21″** have cavity form.
[0090] Advantageously, the elastic retention element **411** has a U shape, and is provided with a convex portion **411′** that projects radially from the lateral surface of the main body **413** and is configured in order to be engaged in the retention recesses **21′**, **21″**.
[0091] Advantageously, the elastic retention element **411** comprises two connection arms **411″**, which connect the convex portion **411′** to the main body **413**, which are configured for being bent during the rotation of the locking pin **41** around the rotation axis R and allow the elastic deformation of the elastic retention element **411**.
[0092] Advantageously, the two connection arms **411″** of the elastic retention element **411** are provided with respective bent portions, which preferably have a U shape, so as to improve the elasticity of the retention element **411**.
[0093] More in detail, the bent portions of the connection arms **411″** of the elastic retention element **411** are susceptible of being compressed and/or enlarged during the rotation of the locking pin **41** around the rotation axis R. In particular, during the rotation of the locking pin **41**, the convex portion **411′** of the elastic retention element **411** will tend to slide against the internal surface **203** of the first housing seat **20**, determining an elastic deformation of the arms **411″** and a consequent compression of a bent portion of a first connection arm **411″** (placed downstream in the rotation sense) and the enlarging of the bent portion of a second connection arm **411″** (placed upstream in the rotation sense).
[0094] Advantageously, the elastic retention element **411** is connected to the main body **413** by

means of only the connection arms **411**″, and preferably lacks any connection at the convex portion **411**′. In other words, the convex portion **411**′ of the elastic retention element **411** is released from other elements, as well as from the connection arms **411**″.

[0095] Advantageously, the main body **413** comprises an enlarged cylindrical base portion **416**, which has an external diameter substantially coinciding with the internal diameter of the first housing seat **20**, so as to determine a housing and a guided rotation, substantially to size, within the first housing seat **20**.

[0096] Advantageously, the cylindrical base portion **416** is extended upwards from the grip portion **415**.

[0097] Advantageously, the aforesaid cylindrical base portion **416** has at least one channel which interrupts the continuity thereof, which is delimited by two lateral walls **417**, and preferably the elastic retention element **411** is interposed between the lateral walls **417** of the cylindrical base portion **416**, with the connection arms **411**″ which project from the lateral walls **417** themselves, as is visible in FIGS. **2-5**.

[0098] Preferably, the convex portion **411**′ of the elastic retention element **411** has a radial bulk that substantially coincides with the external diameter of the cylindrical base portion **416**.

Advantageously, the elastic retention element **411** is placed raised from the grip portion **415**.

Advantageously, the support structure **2** is provided with at least one first stop shoulder **22**, which is placed in the first housing seat **20**, and the locking pin **41** comprises at least one first stop element **412**, which is susceptible of abutting against the first stop shoulder **22** of the first housing seat **20** in order to prevent the rotation of the locking pin **41** in a release sense, opposite the constraint sense.

[0099] Advantageously, the first stop element **412** is an elastic element, which is configured for being elastically deformed at least along a radial direction, substantially orthogonal to the rotation axis R.

[0100] Advantageously, as described above, the locking pin **41** comprises a main body **413**, which is provided with a lateral surface, and the first stop element **412** is extended preferably projecting from the lateral surface of the main body **413**.

[0101] More in detail, the first stop element **412** comprises an elastic tab, which is extended projecting from the lateral surface of the main body **413** up to a free end **412**′ thereof. Preferably, the elastic tab is extended with a substantially L shape.

[0102] In particular, the elastic tab is configured for being bent from an extended position, in which the free end **412**′ of the elastic tab is placed spaced from the lateral surface of the main body **413** and the elastic tab is susceptible of abutting against the first stop shoulder **22** in order to prevent the rotation of the locking pin **41** in the release sense.

[0103] Advantageously, the elastic tab is configured for being bent up to a bent position, in which the free end **412**′ of the elastic tab is placed approached to the lateral surface of the main body **413** in order to allow the free rotation of the locking pin **41** in the constraint sense.

[0104] Advantageously, the support structure **2** is provided with at least one second stop shoulder **23**, which is placed in the first housing seat **20**, and the locking pin **41** comprises at least one second stop element **414**, which is susceptible of abutting against the second stop shoulder **23** of the first housing seat **20** in order to prevent the rotation of the locking pin **41** in the release sense, when the locking pin **41** is in constraint position C.

[0105] Preferably, the second stop element **414** has tooth form, with substantially parallelepiped shape. Advantageously, the second stop element **414** is extended projecting from the lateral surface of the main body **413**, preferably from an opposite side with respect to the first stop element **412**, as is visible in FIG. **15**.

[0106] Advantageously, the head **410** of the locking pin **41** is provided with a first abutment wall **418**, which is directed towards the first opening **200** of the first housing seat **20**.

[0107] More in detail, the head **410** has a parallelepiped form, substantially with rectangular base,

and is provided with a lower face, directed towards the grip portion **415** of the locking pin **41**, and an opposite upper face.
[0108] In particular, when the locking pin **41** is housed in the first housing seat **20**, the lower face of the head **410** is directed towards the first opening **200**.
[0109] Advantageously, the first abutment wall **418** of the head **410** is made on the lower face.
[0110] Advantageously, the support element **3** is provided with a second abutment wall **33**, substantially planar, which is placed in the second housing seat **30** and against which the first abutment wall **418** of the head **410** is susceptible of abutting, in order to constrain the locking pin **41** in the second housing seat **30**.
[0111] More in detail, the second abutment wall **33** of the support element **3** is directed in a sense opposite the first abutment wall **418**.
[0112] Advantageously, the support element **3** is provided with a protuberance, which is extended projecting from an internal surface of the support element **3**, towards the interior of the second housing seat **30**, and which defines the aforesaid second abutment wall **33**.
[0113] Advantageously, the first abutment wall **418** of the head **410** has at least one first section **418′**, tilted with respect to the second abutment wall **33** of the support element **3** by a first angle α, and a second section **418″**, which is tilted with respect to the second abutment wall **33** of the support element **3** by a second angle β, greater than the first angle α.
[0114] Advantageously, with the locking pin **410** in rotation in the constraint sense, the second section **418″** is susceptible of intercepting the first abutment surface **33** of the support element **3**, previously with respect to the first section **418′**.
[0115] More in detail, the second section **418″** of the first abutment wall **418** defines an introduction or slide section, which allows easily rotating the locking pin **410** in the constraint sense when the latter is positioned in the second housing seat **30**, when the head **410** is at the second abutment wall **33** of the support element **3**. In addition, the tilt of the first section **418′** of the first abutment wall **418** determines a forced sliding of the first abutment wall **418** against the second abutment wall **33** of the support element **3** during the rotation of the locking pin **41**, and such forced sliding determines a force oriented substantially orthogonal with respect to the two abutment walls **33**, **418**.
[0116] The aforesaid force determines a movement of the locking pin **41** within the first and the second housing seat **20**, **30**, parallel to the rotation axis R, which is possible due to the normal clearances that are present between the components, and which therefore increases the necessary force for rotating the locking pin **41** in the release sense, opposite the constraint sense.
[0117] In operation, the locking pin **41** is inserted in the first housing seat **20**, by making it translate along an insertion direction X′ substantially parallel to the rotation axis R and to the main extension direction X, in particular by passing the head **410** through the first opening **200**. Subsequently, the pin **41** is rotated around the rotation axis R up to the aforesaid release position B.
[0118] In such release position B, as described above, the support element **3** is released from the support structure **2** and separable from the latter, while the locking pin **41** is advantageously constrained to the support structure **2** along the insertion direction X.
[0119] In other words, when the locking pin **41** is placed in the release position B, it cannot be removed by translating it along the insertion direction X.
[0120] This allows the producer to pre-assemble the locking pin **41** with the support structure **2** and send the support structure **2** thus pre-assembled, reducing the risk of loss of the components (in particular the locking pin **41**) and reducing the number of the necessary operations that the retailer or final client must carry out in order to assemble the chair **1**.
[0121] For such purpose, the support structure **2** is advantageously provided with a retention wall **24**, which is configured for interfering with the locking pin **41** (in particular with its head **410**) in the release position B, in order to prevent the translation of the locking pin **41** along the insertion direction X′.

[0122] More in detail, the retention wall **24** of the support structure **2** is placed substantially parallel to the first abutment wall **418** of the head **410**, and is intended to abuttingly receive such first abutment wall **418**, constraining the head **410**, and thus the locking pin **41**, along the insertion direction X′.

[0123] Advantageously, the retention wall **24** of the support structure **2** is placed in a position diametrically opposite with respect to the locking pin **41**, with respect to the second abutment wall **33** of the support element **3**.

[0124] In particular, the retention wall **24** of the support structure **2** is placed outside the first housing seat **20**, in proximity to the outlet opening **201**. In other words, the locking pin **41** is advantageously inserted in the first housing seat **20** with the head **410** that projects externally from the outlet opening **201**, and is subsequently rotated around the rotation axis R in a manner such to place the head **410** in abutment against the retention wall **24** of the support structure **2**, as is visible in FIG. **6**.

[0125] Also forming the object of the present invention is a method of mounting a chair **1** of the above-described type, regarding which for the sake of description simplicity the same reference numbers will be maintained.

[0126] The method, object of the present invention, comprises a step of arranging the support structure **2**, which is provided with at least one first housing seat **20** provided with the first non-circular opening **200**, and a step of arranging the support element **3**, selected from among a back **31** and a pair of arms **32**, which is provided with the second housing seat **30**.

[0127] The method also comprises a step of inserting the locking pin **41** in the first opening **200** of the first housing seat **20** of the support structure **2**, and a locking step, in which the locking pin **41** is rotated around the rotation axis R in the constraint sense, up to the release position B, in order to retain the locking pin **41** in the first housing seat **20**, mounted in the support structure **2**.

[0128] As set forth above, the insertion step provides for the insertion of the head **410** of the locking pin **41** along the insertion direction X′, until the head **410** does not project outside the outlet opening **201** of the first housing seat **20**, while the central body **413** is housed within the latter. The locking step also provides that the locking pin **41** be rotated around the rotation axis R (in particular in the constraint sense) up to the release position B in order to bring the first abutment wall **418** of the head **410** in abutment against the retention wall **24** of the support structure **2**, constraining the locking pin **41** within the first housing seat **20**, at least along the insertion direction X′. In addition, in such release position B, the locking pin **41** is also advantageously impeded in its rotation around the rotation axis R by the elastic engagement between the elastic retention element **411** and the first retention recess **21′**, in particular with the engagement (preferably shape coupling) between the convex portion **411′** of the elastic retention element **411** and the cavity of the first retention recess **21′**.

[0129] Still more preferably, in such release position B, the locking pin **41** is also prevented in its rotation around the rotation axis R, in the release sense, by the engagement between the second stop element **414** and the second stop shoulder **23** of the first housing seat **20**.

[0130] The method also comprises a step of connecting the support element **3** to the support structure **2**, with the second housing seat **30** aligned and communicating with the first housing seat **20**. More in detail, such connection step provides that the support element **3** be placed with the second housing seat **30** aligned and communicating with the first housing seat **20**, so as to be able to allow the engagement of the locking pin **41** (in particular of the head **410**) in the second housing seat **30**.

[0131] In particular, during the connection step, the back **3** is inserted tilted by an angle such to allow the insertion thereof without interfering with anti-removal elements provided on the support structure **2**, which are visible in FIG. **16**.

[0132] More in detail, as is seen in FIG. **16**, the back **3** has an shaped engagement portion, and the support structure **2** has an insertion cavity that is counter-shaped with respect to the shaped engagement portion of the back **3**, so as to prevent the removal of the back **3** once mounted on the support structure **2**.

[0133] Advantageously, the support element **3** and the support structure **2** are configured for being connected at least partially to each other already in such connection step, preferably by means of a shape coupling of their components, so as to allow the user to mount the support element **3** on the support structure **2** in a nearly stable manner, before then proceeding with the subsequent fixing step without having to manually maintain the support element **3** in position on the support structure **2**. For such purpose, the support element **3** is advantageously provided with an engagement portion that is inserted in a counter-shaped seat of the support structure **2**, retaining the support element **3** in position and preventing the fall thereof during mounting.

[0134] The method also comprises a fixing step, in which the locking pin **41** is rotated around the rotation axis R in the constraint sense, up to the constraint position C, in order to constrain the support structure **2** to the support element **3**.

[0135] More in detail, the fixing step provides that the locking pin **41** be rotated around the rotation axis R up to the constraint position C in order to bring the first abutment wall **418** of the head **410** in abutment against the second abutment wall **33** of the support element **3**, constraining the locking pin **41** within the second housing seat **20** and constraining the support element **3** to the support structure **2**.

[0136] In such constraint position C the locking pin **41** is also impeded in its rotation around the rotation axis R, in the release sense, also by the above-described engagement generated by the tilt between the first abutment wall **418** of the head **410** and the second abutment wall **33** of the support element **3**.

[0137] In addition, in such constraint position C, the locking pin **41** is also advantageously impeded in its rotation around the rotation axis R (in particular in the release sense) by the elastic engagement between the elastic retention element **411** and the second retention recess **21**″, in particular with the engagement (preferably shape coupling) between the convex portion of the elastic retention element **411** and the cavity of the second retention recess **21**″.

[0138] Still more preferably, in such constraint position C, the locking pin **41** is also prevented in its rotation around the rotation axis R, in the constraint sense, by the engagement between the first stop element **412** and the first stop shoulder **22** of the first housing seat **20**.

[0139] The invention thus conceived therefore attains the pre-established objects.

[0140] The contents of the Italian patent application number 102024000005005, from which this application claims priority, are incorporated herein by reference.

Claims

1. A chair for the sitting of a user, said chair comprising: a support structure (**2**), which comprises a support base (**21**), which is intended to be placed in abutment against a ground, and a seat (**210**), which is mounted above said support base (**21**) and is intended to support a user in seated position; wherein said support structure (**2**) is provided with at least one first housing seat (**20**) provided with a first opening (**200**); at least one support element (**3**), which is mechanically connected to said support structure (**2**) and is selected from among a back (**31**) and a pair of arms (**32**), in order to respectively support a back or an arm of the user; wherein said at least one support element (**3**) is provided with a second housing seat (**30**), which is aligned and communicating with the at least one first housing seat (**20**) of said support structure (**2**), when said at least one support element (**3**) is mounted on said support structure (**2**); a removable connection system (**4**), which is placed as a mechanical connection between said support structure (**2**) and said at least one support element (**3**) and comprises a locking pin (**41**) provided with a non-circular head (**410**); wherein said locking pin (**41**) engages said at least one first housing seat (**20**) and said second housing seat (**30**) and is rotatable in a constraint sense, around a rotation axis (R), between at least one release position (B),

in which said locking pin (**41**) releases said at least one support element (**3**) from said support structure (**2**), and a constraint position (C), in which said at least one support element (**3**) is constrained to said support structure (**2**); wherein said locking pin (**41**) comprises an elastic retention element (**411**), and wherein said support structure (**2**) defines, in said at least one first housing seat (**20**), a first retention recess (**21′**), in which said retention element (**411**) is elastically engaged when said locking pin (**41**) is in said constraint position (C), in order to retain said locking pin (**41**) in position, by impeding said locking pin (**41**) from rotating around said rotation axis (R).
2. The chair of claim 1, wherein said support structure (**2**) defines, in said at least one first housing seat (**20**), a second retention recess (**21″**), which is engageable by said elastic retention element (**411**) when said locking pin (**41**) is in said at least one release position (B).
3. The chair of claim 2, wherein said first retention recess (**21′**) and said second retention recess (**21″**) are made on opposite sides of said at least one first housing seat (**20**) and are substantially spaced by an angle equal to 180°.
4. The chair of claim 1, wherein said locking pin (**41**) comprises a main body (**413**), substantially cylindrical, which is provided with a lateral surface; wherein said elastic retention element (**411**) is extended radially projecting from the lateral surface of said main body (**413**); wherein said support structure (**2**) is provided with an internal surface, which is directed towards said at least one first retention seat (**20**) and laterally delimits said at least one first retention seat (**20**); and wherein said first retention access (**21′**) and said second retention recess (**21″**) are made in a depression from the internal surface of said support structure (**2**) along a radial extension direction.
5. The chair of claim 4, wherein said elastic retention element (**411**) has a U shape, wherein said elastic retention element (**411**) is provided with: a convex portion (**411′**) that projects radially from the lateral surface of said main body (**413**) and is configured to be engaged in said first retention recess (**21′**) and second retention recess (**21″**), and two connection arms (**411″**) which connect the convex portion (**411′**) to said main body (**413**), are configured for being bent during rotation of said locking pin (**41**) around said rotation axis (R), and allow elastic deformation of said elastic retention element (**411**).
6. The chair of claim 1, wherein said support structure (**2**) is provided with at least one first stop shoulder (**22**), which is placed in at least one said first housing seat (**20**), and said locking pin (**41**) comprises at least one first stop element (**412**), which is susceptible of abutting against the at least one first stop shoulder (**22**) in order to prevent said locking pin (**41**) from rotating in a release sense, which is opposite said constraint sense.
7. The chair of claim 6, wherein said locking pin (**41**) comprises main body (**413**), substantially cylindrical, which is provided with a lateral surface; wherein said first stop element (**412**) comprises an elastic tab, which is extended projecting from the lateral surface of said main body (**413**) up to a free end (**412′**); wherein said elastic tab is configured for being bent between: am extended position, in which the free end (**412′**) of said elastic tab is placed spaced from the lateral surface of said main body (**413**) and said elastic tab is susceptible of abutting against said first stop shoulder (**22**) in order to prevent said locking pin (**41**) from rotating in said release direction; and a bent position, in which the free end of said elastic tab is placed approached to the lateral surface of said main body (**413**) in order to allow free rotation of said locking pin (**41**) in said constraint sense.
8. The chair of claim 1, wherein said support structure (**2**) is provided with at least one second stop shoulder (**23**), which is placed in said at least one first housing seat (**20**), and wherein said locking pin (**41**) comprises at least one second stop element (**414**), which is susceptible of abutting against the at least one second stop shoulder (**23**) in order to prevent said locking pin (**41**) from rotating in said constraint sense, when said locking pin (**41**) is in said constraint position (C).
9. The chair of claim 1, wherein: said locking pin (**41**) has a substantially L shape, wherein said non-circular head (**410**) is extended substantially orthogonal to said rotation axis (R) and is provided with a first abutment wall (**418**), which is directed towards the at least one first opening

(**200**) of said at least one first housing seat (**20**); said at least one support element (**3**) being provided with a second abutment wall (**33**), substantially planar, which is placed in said second housing seat (**30**) and against which the first abutment wall (**418**) of said non-circular head (**410**) is susceptible of abutting, in order to constrain said locking pin (**41**) in said second housing seat (**30**); wherein the first abutment wall (**418**) of said non-circular head (**410**) has: at least one first section (**418′**), which is tilted with respect to the second abutment wall (**33**) of said at least one support element (**3**) by a first angle (α), and a second section (**418″**), which is tilted with respect to the second abutment wall (**33**) of said at least one support element (**3**) by a second angle (β), which is greater than said first angle (α); wherein, when said locking pin (**410**) is in rotation in said constraint sense, said second section (**418″**) is susceptible of intercepting the second abutment wall (**33**) of said at least one support element (**3**), previously with respect to said at least one first section (**418″**).

10. A method of mounting the chair of claim 1, wherein the method comprises: a step of arranging said support structure (**2**) provided with said at least one first housing seat (**20**) provided with said non-circular first opening (**200**); a step of arranging said at least one support element (**3**), selected from among a back (**31**) and a pair of arms (**32**) and provided with said second housing seat (**30**); a step of inserting said locking pin (**41**) in the at least one first opening (**200**) of the at least one first housing seat (**20**) of said support structure (**2**); a locking step, in which said locking pin (**41**) is rotated around said rotation axis (R) in said constraint sense, up to said at least one release position (B), in order to retain said locking pin (**41**) in said at least one first housing seat (**20**) and mounted in said support structure (**2**); a step of connecting said at least one support element (**3**) to said support structure (**2**), with said second housing seat (**30**) aligned and communicating with said at least one first housing scat (**20**); a fixing step, in which said locking pin (**41**) is rotated around said rotation axis (R) in said constraint sense, up to said constraint position (C), in order to constrain said support structure (**2**) and said at least one support clement (**3**).